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Packet Diagram (v11.0.0+)

Introduction

A packet diagram is a visual representation used to illustrate the structure and contents of a network packet. Network packets are the fundamental units of data transferred over a network.

Usage

This diagram type is particularly useful for developers, network engineers, educators, and students who require a clear and concise way to represent the structure of network packets.

Syntax

```
packet
start: "Block name" %% Single-bit block
start-end: "Block name" %% Multi-bit blocks
... More Fields ...
```

Bits Syntax (v11.7.0+)

Using start and end bit counts can be difficult, especially when modifying a design. For this we add a bit count field, which starts from the end of the previous field automatically. Use `+<count>` to set the number of bits, thus:

```
packet
+1: "Block name" %% Single-bit block
+8: "Block name" %% 8-bit block
9-15: "Manually set start and end, it's fine to mix and match"
... More Fields ...
```

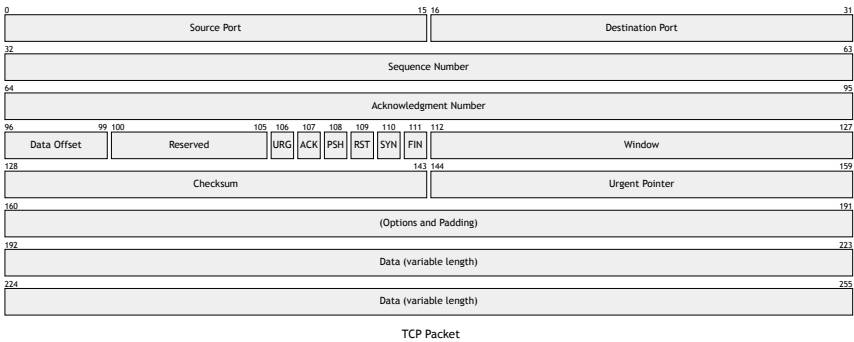
Examples

Code:

```
mermaid
graph TD
    title: "TCP Packet"
    packet
    0-15: "Source Port"
```

```
16-31: "Destination Port"
32-63: "Sequence Number"
64-95: "Acknowledgment Number"
96-99: "Data Offset"
100-105: "Reserved"
106: "URG"
107: "ACK"
108: "PSH"
109: "RST"
110: "SYN"
111: "FIN"
112-127: "Window"
128-143: "Checksum"
144-159: "Urgent Pointer"
160-191: "(Options and Padding)"
192-255: "Data (variable length)"
```

⌘ + Enter | [Run ▶](#)

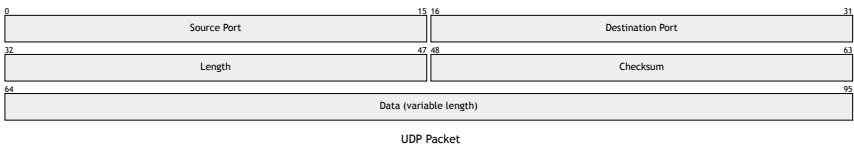


Code:

mermaid

```
packet
title UDP Packet
+16: "Source Port"
+16: "Destination Port"
32-47: "Length"
48-63: "Checksum"
64-95: "Data (variable length)"
```

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Details of Syntax

- **Ranges:** Each line after the title represents a different field in the packet. The range (e.g., 0-15) indicates the bit positions in the packet.
- **Field Description:** A brief description of what the field represents, enclosed in quotes.

Configuration

Please refer to the [configuration](#) guide for details.

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